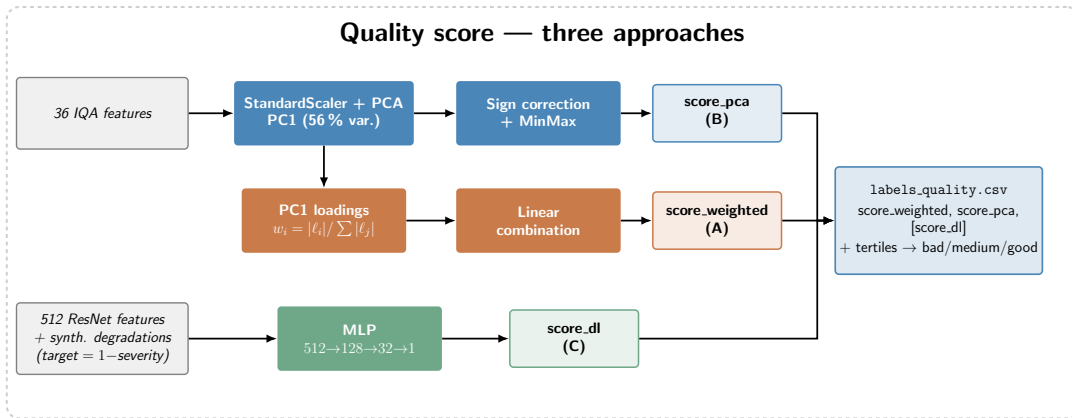
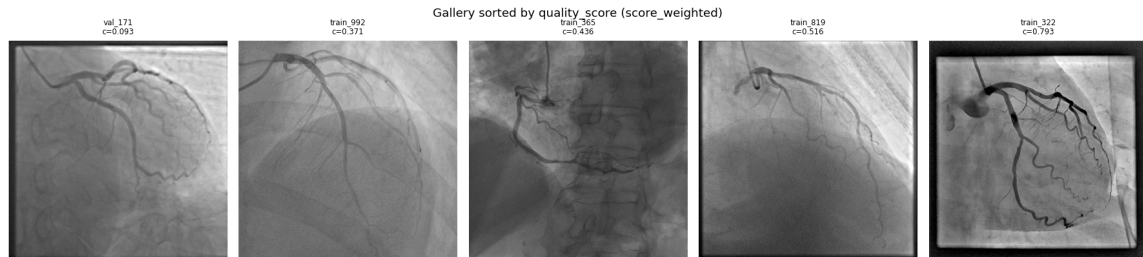


- MedVAE applies the **same transform** to a clean or a heavily degraded coronary angiography: no notion of *how degraded* the input is.
- **Hypothesis:** conditioning MedVAE on a quality score $c \in [0, 1]$, injected via FiLM, should let it adapt to the degradation level actually present.
- Three independent ways to compute c (A: weighted, B: PCA, C: learned) $\rightarrow$ run the *same* pipeline three times.

Building the quality score c

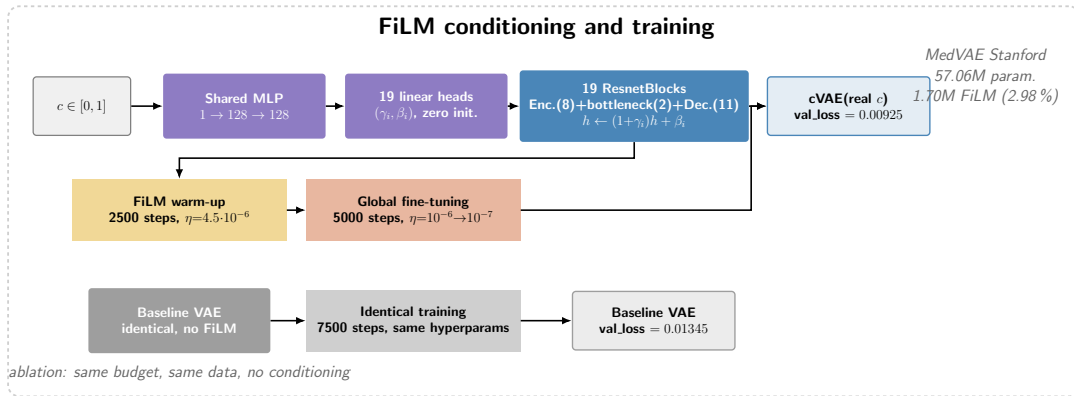


Does the score make clinical sense?

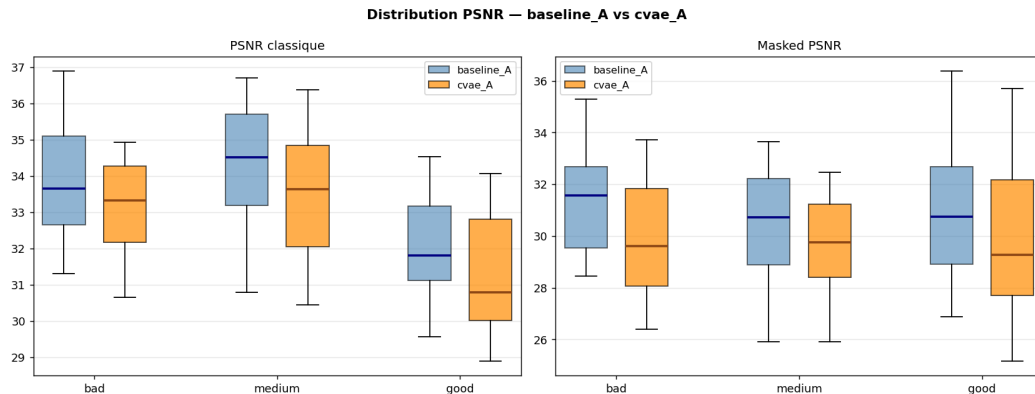


ARCADE images sorted by increasing score c (approach A).

FiLM conditioning & training



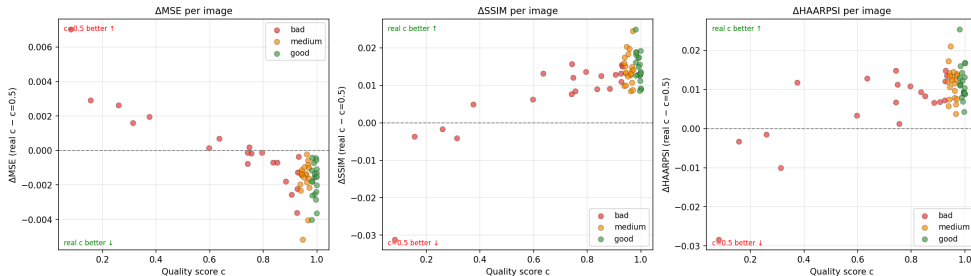
Result: conditioning hurts reconstruction



- Baseline $>$ cVAE on **all 3 approaches** $\times$ **all 4 metrics** (MSE, SSIM, PSNR, HaarPSI), even masked

Ablation: does the network actually use c ?

Per-image difference: cVAE(real c) vs cVAE($c=0.5$) — Approach C



Real c vs. constant $c=0.5$ (approach C): gap grows with image quality: the network's response genuinely depends on c .